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The reaction of iso-propyl radicals with oxygen atoms: Rate coefficient, product branching, and relevance for combustion modeling

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Abstract

Reactions of hydrocarbon radicals with oxygen atoms are important in combustion, and both the rate coefficient and product branching have to be known for an accurate combustion modeling. In this work, the primary product formation in the reaction of the simplest open-chain secondary alkyl radical, iso-propyl (2-C_3H_7), with oxygen atoms in the gas phase was studied at room temperature and a pressure of 4 mbar. 2-C_3H_7 radicals were generated from diisopropylketone ($(2\text{-}C_3H_7)_2CO$) and isopropyliodide (2-C_3H_7I), and O atoms were produced from SO_2 by laser-flash photolysis at $\lambda = 193$ nm, respectively. The reactants and products were detected by quantitative FTIR spectroscopy. The combined product analysis in the experiments with the different precursors leads to the following relative branching fractions: 2-$C_3H_7 + O \rightarrow CH_3CHO + CH_3$ (40%), $CH_3COCH_3 + H$ (36%), $C_3H_6 + OH$ (24%). The channel branching of the iso-propoxy (2-C_3H_7O) radical formed from the 2-$C_3H_7 + O$ reaction was modeled using statistical rate theory with molecular and transition state data from G3MP2B3 calculations. The absolute rate of reaction was studied at room temperature and a pressure of 5.8 mbar. Laser-induced fluorescence (LIF) was used for the specific detection of the OH ($v = 0,1$) radicals, and the rate coefficient of the 2-$C_3H_7 + O$ reaction was derived from the OH ($v = 1$) LIF-time profile leading to $k(2\text{-}C_3H_7 + O) = (1.14 \pm 0.15) \times 10^{14}$ $cm^3/(mol{\cdot}s)$ at 298 K. The OH-forming direct abstraction route and the channel to $CH_3CHO + CH_3$ may influence the flame speed of a propane flame. This is revealed, when the rate coefficient and channel branching of the 2-$C_3H_7 + O$ reaction is incorporated in a suitable detailed reaction mechanism and target experiments are modeled in absence and presence of the title reaction.
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Keywords: Chemical kinetics; Iso-propyl; Statistical rate theory; Combustion modeling

1. Introduction

The main features of the reactions of open-chain primary alkyl radicals (R) such as C_2H_5, 1-C_3H_7, and iso-C_4H_9 with oxygen atoms can be represented by the following scheme (see Refs. [1–6] and references cited therein):

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$$R + O \rightarrow RO^* \quad (1)$$

$$RO^* \rightarrow \text{stable } \beta\text{-C-C scission product} + \text{alkyl} \quad (2)$$

$$RO^* \rightarrow \text{stable } \beta\text{-C-H scission product} + H \quad (3)$$

$$R + O \rightarrow OH + \text{alkene} \quad (4)$$

Numerous experimental and theoretical studies of the reaction of C_2H_5, the simplest alkyl radical bearing both C–C and C–H bonds, with oxygen atoms brought some specific features to light: The branching ratio of the β-C–C and β-C–H scission in the highly excited intermediate $C_2H_5O^*$ is not simply determined by the bond dissociation energies. The absence of OH formation via the isomerization/decomposition path ($O + C_2H_5 \rightarrow C_2H_5O^* \rightarrow C_2H_4OH^* \rightarrow C_2H_4 + OH$) can be explained theoretically. Experimentally it is supported by the detection of OH (vib, $v = 0,1,2,...$) in a non-thermal population of the vibrational states. By determining kinetic data for $C_2H_5 + O$ relative to $CH_3 + O$, both the relative fractions of the main products and the rate coefficients were confirmed [7].

Besides the general mechanistic aspects, alkoxy radicals are essential intermediates in low-temperature and high-temperature combustion of hydrocarbons and atmospheric chemistry as well (see e.g., Refs. [8–12]). One objective in this context is the derivation of simple structure-reactivity correlations which are useful in the modeling of hydrocarbon oxidation.

In contrast to primary alkyl radicals, only few works report on the rates and the mechanisms of secondary alkyl radical + O reactions [13–17]. Here, reactions of cyclic alkyl radicals were in the focus of more recent studies [14,17]. Only few data are available e.g., for the reaction of the 2-C_3H_7, the simplest secondary alkyl radical, with O atoms [13,15,16]. For establishing structure-reactivity correlations, however, more information is needed for secondary alkyl radical + O reactions. Moreover, comparatively stable alkyl radicals with a reactive β-C–H bond are produced during the combustion of cyclic compounds by attack of atoms (H, O) and radicals (CH_3, OH). Further H abstraction from these radicals is assumed to initiate the main path of benzene and soot formation in cyclohexane flames [18]. Consequently, a detailed knowledge of the relative significance of the direct abstraction route (4) to the alkoxy formation route (1) is mandatory to assess the importance of secondary alkyl radical + O reactions these mechanisms.

The present study aims at providing quantitative kinetic data for the reaction

$$2\text{-}C_3H_7 + O \rightarrow \text{products} \quad (5)$$

as the simplest example of a secondary alkyl radical + O reaction to fill the existing gap of knowledge in this reaction class. To this end, the general approach established in preceding papers [5–7,19,20] is followed. We present experimental results on the absolute rate coefficient and the channel branching of reaction (5) as well as theoretical results on the unimolecular decomposition of the chemically activated 2-$C_3H_7O^*$ radical. Under our conditions, the main reaction channels are

$$2\text{-}C_3H_7 + O \rightarrow OH + C_3H_6 \quad (\text{direct H abstraction}) \quad (6)$$

$$2\text{-}C_3H_7 + O \rightarrow 2\text{-}C_3H_7O^* \quad (7)$$

$$2\text{-}C_3H_7O^* \rightarrow CH_3CHO + CH_3 \quad (\beta\text{-C–C scission}) \quad (8)$$

$$2\text{-}C_3H_7O^* \rightarrow CH_3COCH_3 + H \quad (\beta\text{-C–H scission}) \quad (9)$$

$$2\text{-}C_3H_7O^* \rightarrow CH_3C(OH)CH_3 \quad (1,2\text{-H shift}) \quad (10)$$

$$2\text{-}C_3H_7O^* \rightarrow CH_2CH(OH)CH_3. \quad (1,3\text{-H shift}) \quad (11)$$

Finally, we examine and discuss the influence of the title reaction on real combustion systems by implementing the kinetic data in a detailed reaction mechanism [21].

2. Experimental

Two different arrangements were used: The reaction channel yields were studied by arrangement (A) consisting of a laser-flash photolysis setup directly coupled to a FTIR spectrometer for the identification and quantification of stable products. Arrangement (B) is a UV-laser photolysis cell with LIF detection of OH radicals for determining the absolute rate coefficient. Detailed descriptions of the experimental setups are given in Refs. [5,7,19].

2.1. Arrangement (A)

2-C_3H_7 radicals were generated by photolysis of different precursors, $(2\text{-}C_3H_7)_2CO$ and 2-C_3H_7I, at $\lambda = 193$ nm. Oxygen atoms were produced by co-photolysis of SO_2 at the same wavelength. The consumption of the iso-propyl precursors was determined, and all carbon-containing substances were quantified for a mass balance. A quantitative identification was achieved by comparison with appropriately scaled spectra of pure compounds, leading to a total carbon mass recovery of >90% in all experiments. For a detailed description of the calibration procedures see Ref. [7].

2.2. Arrangement (B)

In the UV-laser photolysis cell, co-photolysis of $(2\text{-}C_3H_7)_2CO$ and SO_2 was used for reactant generation. The temporal profile of the OH radical in the ground and the first excited vibrational state was followed by LIF using the pump-probe

technique described in Ref. [19]. The $A^2\Sigma^+(v' = 1, J' = 2\frac{1}{2}) \leftarrow X^2\Pi(v' = 1, J' = 2\frac{1}{2})$ transition of the OH radical was chosen to minimize additional fluorescence from other species (e.g. SO_2) and was saturated to reduce the dependence of LIF intensity on the laser fluence.

2.3. Chemicals

Chemicals were used without further purification:

He (⩾99.995%), Ar (⩾99.998%), SO_2 (⩾99.98%), all Air Liquide; CH_3COCH_3 (⩾99%), $(2\text{-}C_3H_7)_2CO$ (⩾99%), CH_3CHO (⩾99%), $2\text{-}C_3H_7I$ (⩾99%), Fluka; C_3H_6 (⩾99.95%), Linde.

3. Results and discussion

3.1. Overall rate coefficient

The rate coefficient k_5 for the overall reaction (5) was measured under pseudo-first-order conditions ($[O]_0 >> [2\text{-}C_3H_7]_0$) by recording OH LIF-time profiles using arrangement (B). The procedure was introduced in a previous study [5]. It allows the derivation of k_5 from characteristics of the OH time profile independent of the absolute OH (v = i) concentrations and channel fractions ($k_{6,v=i}/k_5$). In the absence of secondary reactions, the OH concentration in the state v = i can be derived by a simple integration of the rate law, leading to

$$[OH](v = i) = [2\text{-}C_3H_7]_0(k_{6,v=i}/k_5)\{1\text{-}\exp[-k_5[O]_0 \cdot t]\} \quad (12)$$

For short reaction times ($t \to 0$), this results in a linear increase with time

$$[OH](v = i)|_{t\to 0} = k_{6,v=i} \cdot [O]_0 \cdot [2\text{-}C_3H_7]_0 \cdot t, \quad (13)$$

and at long reaction times ($t \to \infty$) in a plateau

$$[OH](v = i)|_{t\to\infty} = (k_{6,v=i}/k_5) \cdot [2\text{-}C_3H_7]_0. \quad (14)$$

By equating Eqs. (13) and (14), the rate coefficient k_5 can be deduced from the time t_{in}, where the straight lines from the two limiting cases intercept, in a very good approximation by

$$k_5 = 1/([O]_0 \cdot t_{in}). \quad (15)$$

Figure 1 shows the OH ($v = 1$) time profile, and the straight lines through initial and final concentrations intercepting at t_{in}. The validity of this approach was confirmed previously by simulating OH profiles for different reaction conditions and alkyl radical precursors [5]. For the overall rate coefficient of the title reaction, we obtain

$$k_5 = (1.14 \pm 0.15)10^{14}\ cm^3/(mol \cdot s) \quad (16)$$

at $T = 298$ K and $P = 5.8$ mbar. The given error margin was estimated from the combined uncertainties in the determination of the SO_2 concentrations, the absorption cross-section, and the time t_{in}. The statistical error is well below this 15% limit. OH ($v = 0$) was also detected, but the kinetic evaluation was based on OH ($v = 1$) because of the better signal-to-noise ratio.

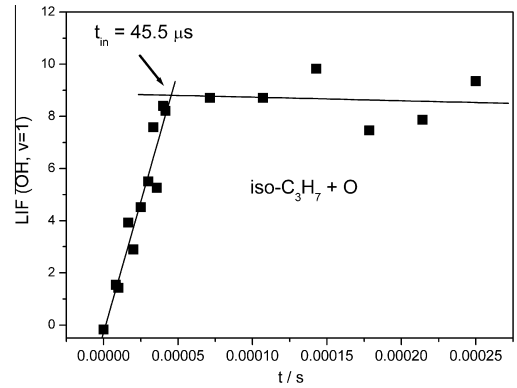

Fig. 1. Determination of the rate coefficient k_5: OH ($v = 1$) LIF-time profile, $t_{in} = 45.5$ µs (see text). $p = 5.8$ mbar, mole fractions: $x((2\text{-}C_3H_7)_2CO) = 0.084$ mol%, $x((SO_2) = 0.75$ mol%, rest: He, laser energy = 9.8 mJ/pulse, $[O]_0 = 2.03 \times 10^{-10}$ mol/cm^3 is derived from $[SO_2]_0$, laser fluence and SO_2 absorption cross-section at 193 nm (252 m^2/mol, base 10, see Ref. [5]).

In the literature, only one experimental value $k_5 = 1.32 \times 10^{14}$ cm^3/(mol·s) is reported [22], which is in reasonable agreement with the present study and the estimation of 9.64×10^{13} cm^3/(mol·s) by Tsang [15]. Another estimated value of 3×10^{14} cm^3/(mol·s) by Herron [16] is too high, which is a similar finding compared to the reaction $1\text{-}C_3H_7 + O$ [5]. Previous experimental studies on other open-chain alkyl + O reactions (see e.g. Refs. [2] and [20]) suggest that the rate coefficient of $2\text{-}C_3H_7 + O$ shows no or only a minor temperature dependence.

3.2. Experimental determination of the product branching

The formation of the stable products was studied by using arrangement (A), where the iso-propyl radicals were generated by photodecomposition of the precursors $(2\text{-}C_3H_7)_2CO$ or $2\text{-}C_3H_7I$. Product spectra were taken in intervals of 50 laser shots each up to a total number of 200 laser shots; this number corresponds to a total precursor conversion of 10%, which is an upper limit in order to avoid secondary reactions.

The IR spectra were analyzed in the spectral region of 650–3800 cm^{-1} by comparing the product spectra with spectra of the pure substances. IR spectra and calibration curves were recorded and determined for all identified compounds at the same total pressure and temperature (4 mbar

and 298 K). Due to their low molecular masses and their different functional groups (–CHO, –C=C– and –C–CO–C–) absorbing in different spectral regions, an unambiguous and quantitative calibration of the major products was possible.

An example for a unique identification is given in Fig. 2 for the precursor $(2\text{-}C_3H_7)_2CO$. In separate spectral regions an unambiguous identification of the following stable products was achieved: acetaldehyde (CH_3CHO, reaction (8)), acetone (CH_3COCH_3, reaction (9)) and propene (C_3H_6, reaction (6)). For determining the channel branching, we averaged the integrated IR spectra by plotting product yields as a function of consumed precursor. The relative slopes of the straight lines through the origin are assigned to the relative product yields as shown in Fig. 3. We found that acetaldehyde resulting from β-C–C scission (reaction (8)) is the main product with a slightly higher yield than the β-C–H scission product acetone (reaction (9)). This finding is not influenced by different ratios of initial $2\text{-}C_3H_7$ and O atom concentrations and different $2\text{-}C_3H_7$ precursors. The effect of initial reactant concentrations was tested by varying precursor concentrations such that either $2\text{-}C_3H_7$ (0.4 mbar $(2\text{-}C_3H_7)_2CO$, 0.05 mbar SO_2) or O atoms (0.4 mbar $(2\text{-}C_3H_7)_2CO$, 0.2 mbar SO_2) were in excess after photodecomposition. In these experiments and by using $2\text{-}C_3H_7I$ as iso-propyl precursor, the derived product distributions were constant with respect to the uncertainty of the calibration procedure. Therefore, we can conclude that under our experimental conditions, the primary reaction is kinetically well separated from secondary chemistry allowing the determination of channel fractions from the stable primary products. This is in line with the findings in a series of preceding studies [5–7,14].

In contrast to the precursor $2\text{-}C_3H_7I$, where $HI + C_3H_6$ is a main photodecomposition channel at 193 nm, propene is not formed in significant amounts from $(2\text{-}C_3H_7)_2CO$ photolysis at this wavelength. Therefore, we could determine the yield of reaction (6), the direct H atom abstraction, from the propene concentrations. Compared to the main reaction channel (8), we found a relative ratio of 0.6 for the H atom abstraction from $2\text{-}C_3H_7$ (see Fig. 3) independent of initial reactant concentrations. The in situ LIF detection of OH

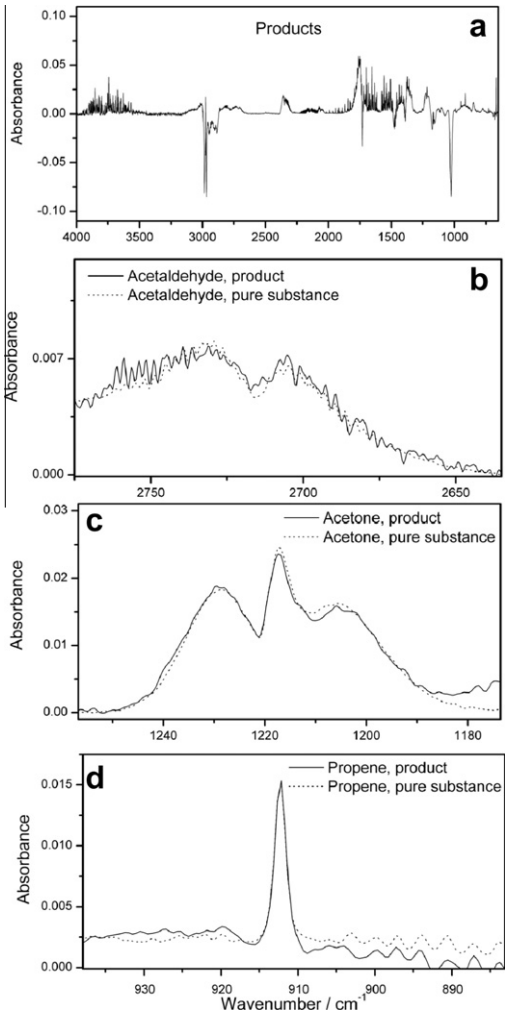

Fig. 2. Identification of major products (precursor $(2\text{-}C_3H_7)_2CO$), (a) product spectrum, (b) acetaldehyde (β-C–C scission product) (c) acetone (β-C–H scission product), (d) propene; $p((2\text{-}C_3H_7)_2CO) = 0.4$ mbar, $p(SO_2) = 0.2$ mbar, p(total) = 4 mbar, bath gas Ar, $T = 298$ K.

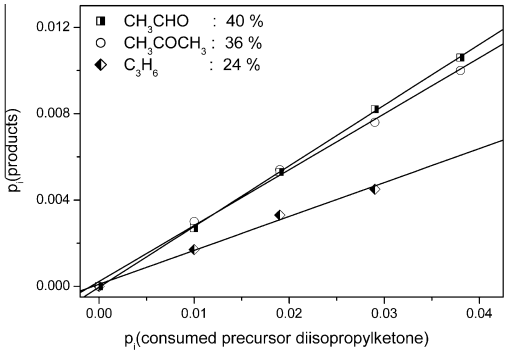

Fig. 3. Formation of products vs. consumption of precursors, $p((2\text{-}C_3H_7)_2CO) = 0.4$ mbar, $p(SO_2) = 0.1$ mbar, p(total) = 4 mbar, bath gas: Ar, $T = 298$ K. $[2\text{-}C_3H_7]_0$ and $[O]_0$ can be estimated from absorption cross-sections at 193 nm (SO_2: 252 m^2/mol, base 10, $(2\text{-}C_3H_7)_2CO$: ca. 50 m^2/mol, base 10), the laser fluence of 17 mJ/cm^2 and the yield of ca. 1.4 iso-propyl radicals per $(2\text{-}C_3H_7)_2CO$ being photolyzed.

showed that more OH is formed in $v = 1$ than in the ground state confirming the direct abstraction mechanism. In summary, we found three major channels for the reaction of the secondary alkyl radical 2-C_3H_7 with O atoms in accordance with the established mechanism for the analogous reactions of primary radicals: 76% of the the iso-propyl radicals react via the iso-propoxy radical (yields: β-C–C: 40%, β-C–H: 36%) and 24% via H abstraction forming directly propene + OH. In Table 1, the experimental results are summarized and compared with theoretical predictions as discussed below.

There exists only one experimental study in the literature on the iso-propyl + O reaction [13] reporting a branching ratio β-C–C/β-C–H of 1, which is in good agreement with our results. The abstraction channel could not be quantified in the early work [13]. Interestingly, the iso-propyl + O reaction shows kinetics very similar to the analogous reaction of the isomer *n*-propyl (1-C_3H_7) [5]. Both the branching ratio β-C–C/β-C–H (1.1:1 (2-C_3H_7) and 1.4:1 (1-C_3H_7)) and the absolute fraction of the abstraction route (24%) are similar or identical [5]. Comparing the branching ratio β-C–C/β-C–H for iso-propoxy (2-C_3H_7O) with the smaller primary ethoxy (C_2H_5O) and larger secondary cyclic radicals, we see an increase with the alkoxy radical size: β-C–C scission is the minor channel for the chemically activated ethoxy radical (1:1.4) [2,4,5], competes for the two propoxy radicals and is the major channel for cyclic alkoxy radicals (≈5:1 for c-C_5H_9O, c-$C_6H_{11}O$, c-$C_8H_{15}O$) [14,17].

The situation changes for the mechanistically different direct H atom abstraction channel. Here a constant channel fraction of around 25% emerges as a general feature of the reaction of O atoms with saturated, non-cyclic alkyl radicals [2,5,6]. The constant fraction of H abstraction for the saturated C2–C4 alkyl radicals examined so far is surprising since other structure modifications strongly influence the fraction of H atom abstraction: By introducing a double bond as in C_3H_5, the fraction is suppressed to below 5% [20], while for cyclic alkyl radicals a strong variation is found: it changes with the ring size between 35% for c-C_5H_9 [14] and <5% for c-C_6H_{11} to 11% for c-C_8H_{15} [17]. These findings point to the difficulties arising when reaction mechanisms for cyclic hydrocarbons [18] or intermediate soot precursors [23] are compiled purely on the basis of established kinetic data for open-chain species. In the present study we did not examine the pressure dependence of the main product channels. From previous studies (e.g. Ref. [5], [20]) it is experimentally and theoretically established that up to 1 bar for small alkyl radicals (less than 6 C atoms) the channel branching is not pressure dependent in alkyl + O reactions. This picture changes for larger cyclic alkyl radicals [17].

3.3. Modeling of the product branching

The geometries, rotational constants and harmonic vibrational frequencies (scaled by 0.961 [24]) for iso-propoxy and the transition states (TS) for reactions (8)–(11) were calculated with density functional theory (DFT [25]) at B3LYP/6-31G(d) level of theory [26,27]. The electronic energies were obtained with the G3MP2B3 method [28]. All quantum chemical calculations were performed with the Gaussian03 program suite [29].

Stationary points of the potential energy surface calculated with the G3MP2B3 method are depicted in Fig. 4. The β-C–C scission, reaction (8), is predicted to be the lowest-energy decomposition channel with a threshold energy of $E_{0,8} = 55.8$ kJ/mol, while a value of $E_{0,9} = 70.5$ kJ/mol for the β-C–H scission, reaction (9), is found. For the 1,2-H (10) and 1,3-H (11) shift reactions, threshold energies of 103.7 and 113.0 kJ/mol were obtained, respectively. The higher values of the threshold energies of the latter reactions in addition to the tighter cyclic transition states compared to the scission reactions indicate that H shift reactions are only of minor importance. A critical parameter with respect to the product branching is the difference in the threshold energies for the β-C–C and β-C–H scission reactions, $E_{0,9} - E_{0,8}$, which is calculated to be 14.7 kJ/mol

Table 1
Experimentally determined and theoretically predicted relative branching fractions and calculated low-pressure limiting rate coefficients $k_{0,i}$ of the 2-C_3H_7 + O reaction.

	Branching fractions/%				$k_{0,i}/s^{-1}$
	Experiment			Theory[b]	
Reaction	$(2\text{-}C_3H_7)_2CO$	iso-C_3H_7[a]	Average		
(6)	24	(24)	24 ± 3	–	–
(8)	39	41	40 ± 3	85	$3.03 \cdot 10^{13}$
(9)	37	35	36 ± 3	13	$4.73 \cdot 10^{12}$
(10)	–	–	–	<1	$3.39 \cdot 10^{11}$
(11)	–	–	–	<1	$1.18 \cdot 10^{11}$

[a] A 24% channel fraction of direct H abstraction is assumed.
[b] Abstraction reaction (6) is not included.

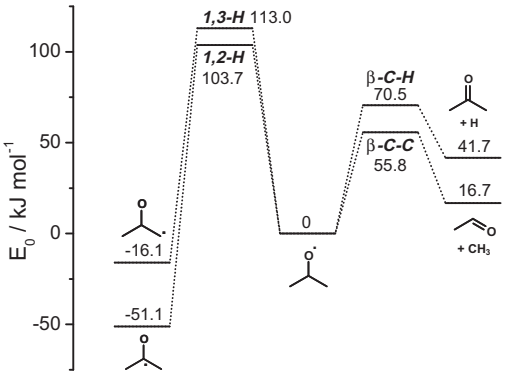

Fig. 4. Potential energy diagram for the unimolecular reactions of 2-C_3H_7O calculated with the G3MP2B3 method. E_0 denotes the energy at 0 K relative to 2-C_3H_7O.

with the G3MP3B3 method. According to Table 2, this value is confirmed by several high-level and compound methods within a range of ≈4 kJ/mol. The employed DFT methods seem to systematically overestimate this value, while at MP2/6-31G(d) level of theory, the threshold energies for reactions (8) and (9) are virtually identical, indicating problems in the MP2 calculations with a small basis set for this system. In a previous study of the thermal decomposition of 2-C_3H_7O, activation enthalpies at $T = 298$ K for reactions (8) and (9) were determined as 61.2 and 76.4 kJ/mol, respectively, with a G2MP2-like procedure based on MP2/6-311G** geometries [30]. Neglecting the influence of thermal contributions, these values are approximately 5 kJ/mol higher in energy than the ones found with the G3MP2B3 method, while the difference (76.4 − 61.2) kJ/mol = 15.2 kJ/mol is in close agreement with the results for $E_{0,9} - E_{0,8}$ in this work.

The product branching is determined from statistical unimolecular rate theory in analogy to Ref. [19]. Under our experimental conditions, the branching fractions for the decomposition channels of 2-C_3H_7O can be obtained from the low-pressure limiting rate coefficients for chemical activation k_0. These can be calculated from the nascent distribution F(E) of 2-C_3H_7O and the specific rate coefficients k(E) according to RRKM theory [31]:

$$k_i(E) = L_i \frac{W_i(E)}{h\rho(E)}, \tag{17}$$

with the reaction path degeneracy L_i [32], the sum of states $W_i(E)$ of the transition state for the i-th channel, respectively, Planck's constant h, and the density of states $\rho(E)$ of 2-C_3H_7O. The calculation of $F(E)$ requires the number of open reaction channels, $W(E - E_0^{ar})$ (see [19]), for the reaction 2-C_3H_7 + O. In contrast to Ref. [19], this quantity is approximated here by the total rovibrational sum of states of 2-C_3H_7, where the three external rotational degrees of freedom were treated as independent one-dimensional rotators. In this way, the loose character of the transition state for the 2-C_3H_7 + O reaction is accounted for. We note that details in the calculation of $W(E - E_0^{ar})$ have only a marginal influence on the branching ratio.

The sums and densities of states were counted on the basis of the harmonic vibrational wavenumbers using the Beyer–Swinehart algorithm for a value of the angular momentum quantum number $J = 26$, which corresponds to the average rotational excitation at $T = 300$ K [33,34]. The 2-C_3H_7O radicals formed from the 2-C_3H_7 + O reaction possess an average initial excess excitation energy of 394 kJ/mol, which is obtained using enthalpy data from the literature [35,36]. For the reaction path degeneracy, we used $L_i = 2$ for reac-

Table 2
Comparison of the threshold energies E_0 at 0 K for the unimolecular reaction channels of 2-C_3H_7O obtained with different quantum chemical methods.

Method	E_0/(kJ/mol)				$(E_{0,9} - E_{0,8})$/(kJ/mol)
	β-C–C (8)	β-C–H (9)	1,2-H (10)	1,3-H (11)	
B3LYP/6-31G(d)[a]	65.4	89.2	122.7	129.2	23.8
B3LYP/cc-pVTZ[b]	55.7	83.3	106.7	115.7	27.6
MP2/6-31G(d)[c]	91.6	92.1	122.1	136.1	0.5
MP2/cc-pVTZ[d]	75.2	87.1	96.8	111.0	11.9
G3MP2B3	55.8	70.5	103.7	113.0	14.7
G3B3	54.2	69.9	101.9	111.0	15.7
G2MP2	55.6	71.3	101.4	112.4	15.7
QCISD(T)/cc-pVDZ[e]	67.3	80.1	117.9	122.2	12.8
QCISD(T)/cc-pVTZ[e]	61.7	81.0	107.1	113.1	19.3
CCSD(T)/cc-pVTZ[e]	63.4	82.7	107.7	113.8	19.3

[a–d]Frequencies scaled by 0.961[a] [24], 0.9691[b] [39], 0.9247[c] [40], 0.9575[d] [39].
[e]Based on geometries and frequencies calculated at B3LYP/6-31G(d) level of theory.

tions (8) and (11), and $L_i = 1$ for reactions (9) and (10).

Specific rate coefficients $k_i(E)$ for reactions (8)–(11) are plotted in Fig. 5 together with the nascent distribution $F(E)$ of the formed 2-C_3H_7O radicals. The resulting low-pressure limiting rate coefficients $k_{0,i}$ and the relative branching ratios are given in Table 1. In agreement with the experimental results, the β-C–C scission is predicted to be the main reaction channel. The calculated branching ratio between the β-C–C and β-C–H scission of 85%/13% ≈ 6.5 exceeds the experimentally determined value of 1.1. The yields of the 1,2-H and 1,3-H shift reaction amount to <1%, respectively, which means that these channels are negligible.

The lifetime of the chemically activated 2-C_3H_7O radicals is predicted to be ≈0.03 ps compared to ≈0.2 ps found for 1-C_3H_7O in an earlier study [5]. For the 1-C_3H_7O radicals, the calculated branching ratio of β-C–C/β-C–H ≈ 2.9 is closer to the experimentally determined value of 1.4 than found in the present work for 2-C_3H_7O (see above). The difference in the calculated branching ratios β-C–C/β-C–H of 6.5 for 2-C_3H_7O and 2.9 for 1-C_3H_7O can be qualitatively understood by the shift of the reaction path degeneracies from $L_i = 2$ (2-C_3H_7O) to $L_i = 1$ (1-C_3H_7O) for the β-C–C scission and from $L_i = 1$ (2-C_3H_7O) to $L_i = 2$ (1-C_3H_7O) for the β-C–H scission. Based on these shifts alone, an increase of the branching ratio β-C–C/β-C–H by a factor of 4 is expected in going from 1-C_3H_7O to 2-C_3H_7O. The actual ratio (β-C–C/β-C–H)(2-C_3H_7O)/(β-C-C/β-C-H)(1-C_3H_7O) of only 2.2 can be explained by the decrease in the difference of the threshold energies E_0(β-C-H) – E_0(β-C–C) from 26.6 kJ/mol for 1-C_3H_7O [5] to 14.7 kJ/mol for 2-C_3H_7O. Increasing the difference E_0(β-C–H) – E_0(β-C–C) for 2-C_3H_7O to 26.6 kJ/mol by raising the threshold energy $E_{0,9}$ for the β-C–H scission from 70.5 to 82.2 kJ/mol, leads to a branching ratio β-C–C/β-C–H of 11.7, which is exactly 4 times the value of 2.9 calculated for 1-C_3H_7O documenting the influence of L_i. The calculated short lifetime of 2-C_3H_7O indicates, that the limit of the applicability of statistical theories is reached and that non-statistical effects might influence the product branching. This might be a reason for the poorer agreement between the calculated and experimentally determined branching ratio β-C–C/β-C–H for 2-C_3H_7O compared to 1-C_3H_7O, where a longer lifetime was observed [5]. To resolve this discrepancy, more work on the kinetics of highly excited species in general and on the decomposition of chemically activated secondary alkoxy radicals is clearly needed.

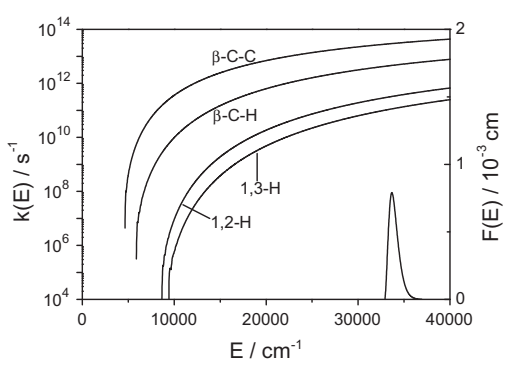

Fig. 5. Specific rate coefficients $k_i(E)$ for the decomposition reactions and nascent distribution F(E) of 2-C_3H_7O.

3.4. Impact on the modeling of specific combustion systems

We tested the impact of the 2-C_3H_7 + O kinetics on combustion systems for two target experiments: the velocity of a free propagating propane flame and the ignition delay times of a stoichiometric propane/oxygen/argon mixture. For the calculations, we used a comprehensive hydrocarbon oxidation mechanism [21], in which the 2-C_3H_7 + O reaction was not originally included. First we tested the original mechanism extended by the 2-C_3H_7 + O reaction with the rate coefficient of the present study. We found that the title reaction had no impact on the ignition delay times from 1000–1800 K, while a slight decrease of 0.05% was found for the flame speed under stoichiometric conditions. To further elucidate the impact of the 2-C_3H_7 + O reaction, the rate coefficients were modified channel-specifically. The result was that only reactions 6 and 8 (H abstraction and β-C–C fission) are sensitive on the flame speeds. This sensitivity can be drastically increased if a lower rate coefficient for the unimolecular decomposition of 2-C_3H_7 to propene + H is assumed. For measured and theoretically predicted rate coefficients of such reactions, high uncertainties have to be assigned, which may amount to more than a factor of ten (see Refs. [15,37] and references cited therein). If we apply a rate coefficient for the iso-propyl decomposition reduced by a factor of ten compared to Ref. [15], a decrease in propane flame speed follows which adds up to 1.5% under fuel-lean conditions upon inclusion of 2-C_3H_7 + O with the kinetic data reported in this study. In this case the title reaction would advance to the most sensitive reactions on propane flame speeds of species carrying more than one C atom (see discussion on alkane flame speeds in Refs. [21,38]). We note here that 1-C_3H_7 + O is not sensitive because 1-C_3H_7 is less stable than 2-C_3H_7.

These examples show that the title reaction may have an impact on propane combustion under flame conditions. Together with the

previous results on $C_3H_5 + O$ [20], we see that hydrocarbon radical + O reactions in general become sensitive if the hydrocarbon radical is stabilized, which is the case for unsaturated radicals such as C_3H_3, C_3H_5 or the cyclic C_5H_5 (cyclopentadienyl).

4. Conclusions

The rate coefficient and product distribution of the 2-$C_3H_7 + O$ reaction were experimentally determined and the channel branching of the chemically activated 2-C_3H_7O radical was calculated from statistical rate theory. Experiment and theory agree in showing that β-C–C scission is the main decomposition channel of the intermediate 2-C_3H_7O radical, while the competing β-C–H scission is underpredicted by theory. This may be related to limits of the statistical description due to the very short lifetime of 2-C_3H_7O of ≈0.03 ps. On the basis of our measurements, we confirm a constant channel fraction of ≈25% for direct H atom abstraction as a general feature for the reaction of O atoms with saturated, noncyclic alkyl radicals. Finally, we examined the influence of the title reaction on real combustion systems by implementing the kinetic data in a detailed reaction mechanism. We found that H atom abstraction from iso-propyl and β-C–C fission of iso-propoxy may influence propane combustion under flame conditions, while β-C–H fission does not.

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